

THE RANKER'S MASTERBOOK

ARC.

The Inverse Circular Functions

The Problem Set

100

PROBLEMS

10

CHAPTERS

3–5

DIFFICULTY

arc

*One hundred original problems on the inverse circular functions – where the branch you forget is the mark you lose.
For the very top of JEE Advanced and the Mathematical Olympiad. Solutions in the companion volume.*

Contents

Ten chapters, ten problems each.

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CHAPTER I

Principal Branches & the Range Trap

$\sin^{-1}$

Outside the principal range, $\arcsin \circ \sin$ is a sawtooth.

№1 Ten Radians Out

● ● ● ● ● L3

EVALUATE

$$\sin^{-1}(\sin 10)$$

sin-inverse

reduction

quadrant

radian-measure

№2 The Negative Decoy

● ● ● ● ● L3

EVALUATE

$$\cos^{-1}(\cos(-5))$$

cos-inverse

even-function

reduction

quadrant

№3 Tangent's Quiet Period

● ● ● ● ● L3

EVALUATE

$$\tan^{-1}(\tan 3)$$

tan-inverse

period-pi

reduction

open-interval

№4 Seven-Sixths of a Pi

● ● ● ● ● L3

EVALUATE

$$\sin^{-1}\left(\sin \frac{7\pi}{6}\right)$$

sin-inverse

exact-angle

third-quadrant

reduction

№5 Cotangent's Wrong Door

● ● ● ● ● L4

EVALUATE

$$\cot^{-1}(\cot(-4))$$

cot-inverse

negative-angle

open-interval-0-pi

period-pi

№6 Eight and the Vanishing Pi

●●●●● L4

EVALUATE

$$\sin^{-1}(\sin 8) + \cos^{-1}(\cos 8)$$

sin-inverse

cos-inverse

sum

complementary-cancellation

№7 Where the Sawtooth Meets the Ray

●●●●● L4

FIND THE NUMBER OF SOLUTIONS

$$\text{the equation } \cos^{-1}(\cos x) = \frac{x}{4} \text{ on } [0, 4\pi]$$

cos-inverse

triangular-wave

equation-counting

graph-intersection

№8 Three Inverses, One Integer

●●●●● L5

EVALUATE

$$\sin^{-1}(\sin 5) + \cos^{-1}(\cos 11) + \tan^{-1}(\tan 7)$$

sin-inverse

cos-inverse

tan-inverse

mixed-sum

pi-cancellation

№9 Slope of the Sawtooth at Seventeen

●●●●● L4

DIFFERENTIATE

$$g(x) = \cos^{-1}(\cos x); \text{ find } g(17) \text{ and } g'(17)$$

cos-inverse

derivative

branch-slope

piecewise

№10 Six Integers Into the Sine Mill

●●●●● L5

EVALUATE

$$\sum_{k=1}^6 \sin^{-1}(\sin k)$$

sin-inverse

integer-arguments

telescoping-pi

quadrant-bookkeeping

CHAPTER II

Addition Formulas & $\pm\pi$ Corrections



Add two arctangents and the formula quietly leaks a π .

№11 The First Overflow

●●●●● L4

EVALUATE

$$\tan^{-1} 2 + \tan^{-1} 3$$

tan-inverse

addition-formula

pi-correction

principal-value

№12 Doubling Past the Line

●●●●● L4

EVALUATE

$$2 \tan^{-1} (2 + \sqrt{3})$$

double-angle

tan-inverse

pi-correction

surd

№13 Mirror of the Overflow

●●●●● L4

EVALUATE

$$\tan^{-1}(-2) + \tan^{-1}(-3)$$

tan-inverse

negative-correction

addition-formula

quadrant

№14 Reciprocal Bookkeeping

●●●●● L4

EVALUATE

$$\tan^{-1} \frac{1}{3} + \tan^{-1} 3 + \tan^{-1} \left(-\frac{1}{5}\right) + \tan^{-1}(-5)$$

reciprocal-identity

tan-inverse

sign-of-x

telescoping

№15 Crossing the Sine Boundary

●●●●● L4

EVALUATE

$$\sin^{-1} \frac{4}{5} + \sin^{-1} \frac{12}{13}$$

sin-inverse

boundary-crossing

pythagorean

supplement

№16 Two Obtuse Cosines

●●●●● L4

EVALUATE

$$\cos^{-1}\left(-\frac{1}{2}\right) + \cos^{-1}\left(-\frac{1}{\sqrt{2}}\right)$$

cos-inverse

addition-formula

2pi-correction

sum-sign

№17 The Möbius Argument

●●●●● L4

SIMPLIFY

$$f(x) = \tan^{-1}\left(\frac{1+x}{1-x}\right) \quad \text{for } x > 1$$

tan-inverse

mobius

pi-correction

branch-condition

№18 The Extraneous Root

●●●●● L4

SOLVE FOR X

$$\tan^{-1}(2x) + \tan^{-1}(3x) = \frac{\pi}{4}$$

tan-inverse

equation

extraneous-root

pi-correction

№19 Counting Honest Solutions

●●●●● L4

FIND THE NUMBER OF SOLUTIONS

$$\tan^{-1}(x+1) + \tan^{-1}(x-1) = \tan^{-1}\frac{8}{31}, \quad x \in \mathbb{R}$$

tan-inverse

solution-count

extraneous-root

monotonicity

№20 Two Overflows Cancel

●●●●● L4

PROVE THAT

$$2 \tan^{-1} 2 + \tan^{-1} \frac{4}{3} = \pi$$

tan-inverse

double-angle

pi-correction

telescoping

CHAPTER III

Telescoping Arctangents



Each term is a difference of two angles – the sum collapses.

№21 The Gauss Cascade

● ● ● ● ● L3

EVALUATE

$$\sum_{k=1}^{\infty} \tan^{-1} \left(\frac{1}{k^2 + k + 1} \right)$$

telescoping

infinite-series

difference-formula

principal-value

№22 Odd Steps, Even Squares

● ● ● ● ● L3

PROVE THAT

$$\sum_{k=1}^n \tan^{-1} \left(\frac{1}{2k^2} \right) = \tan^{-1}(2n+1) - \frac{\pi}{4} \quad (n \geq 1)$$

telescoping

finite-sum

induction-free

difference-formula

№23 The Skip-a-Step Sum

● ● ● ● ● L4

EVALUATE

$$\sum_{k=1}^{\infty} \tan^{-1} \left(\frac{2}{k^2} \right)$$

telescoping

two-step

infinite-series

boundary-terms

№24 A Parameter That Forgets Its Sign

● ● ● ● ● L4

EVALUATE

$$S(x) = \sum_{k=0}^{\infty} \tan^{-1} \left(\frac{x}{1 + k(k+1)x^2} \right), \quad x \in \mathbb{R}$$

telescoping

parametric

sign-trap

infinite-series

№25 Squares Within Squares

● ● ● ● ● L4

EVALUATE

$$\sum_{k=1}^{\infty} \tan^{-1} \left(\frac{2k+1}{1 + k^2(k+1)^2} \right)$$

telescoping

perfect-squares

infinite-series

difference-formula

№26 Start at Two, Land on a Third

●●●●● L4

EVALUATE

$$\sum_{k=2}^{\infty} \tan^{-1} \left(\frac{2k}{k^4 + k^2 + 2} \right)$$

telescoping shifted-index biquadratic factorisation

№27 The Ladder to a Right Angle

●●●●● L4

EVALUATE

$$\sum_{k=1}^{\infty} \tan^{-1} \left(\frac{1}{k^2 - k + 1} \right)$$

telescoping infinite-series half-pi difference-formula

№28 Halving the Odd Reciprocals

●●●●● L4

EVALUATE

$$\sum_{k=1}^{\infty} \tan^{-1} \left(\frac{4}{4k^2 + 3} \right)$$

telescoping infinite-series odd-denominators closed-form

№29 An Irrational Step Size

●●●●● L5

EVALUATE

$$\sum_{k=1}^{\infty} \tan^{-1} \left(\frac{\sqrt{2}}{2k^2 + 2k + 1} \right)$$

telescoping surd infinite-series co-function principal-value

№30 Two Constants Escape

●●●●● L5

EVALUATE

$$\sum_{k=1}^{\infty} \tan^{-1} \left(\frac{4}{4k^2 - 3} \right)$$

telescoping two-step boundary-terms infinite-series branch-care

CHAPTER IV

Identities & Collapse



Most nested inverse-trig expressions are secretly a straight line.

№31 The Sign Trap at Three-Quarters

● ● ● ● ● L3

EVALUATE

Evaluate $\sin^{-1}(2x\sqrt{1-x^2})$ at $x = \sin \frac{5\pi}{12}$.

sine-double-angle

branch

principal-value

collapse

№32 Piecewise or Bust

● ● ● ● ● L4

SIMPLIFY

Simplify $f(x) = \sin^{-1}(2x\sqrt{1-x^2})$ on $[-1, 1]$, giving the full piecewise form with domains.

sine-double-angle

piecewise

domain

collapse

№33 Triple Angle Overshoot

● ● ● ● ● L3

EVALUATE

Evaluate $\tan^{-1}\left(\frac{3x-x^3}{1-3x^2}\right) - 3\tan^{-1}x$ at $x = 2$.

tangent-triple-angle

branch

collapse

principal-value

№34 Where Three Times Holds

● ● ● ● ● L4

SOLVE FOR X

Find all x for which $\tan^{-1}\left(\frac{3x-x^3}{1-3x^2}\right) = 3\tan^{-1}x$.

tangent-triple-angle

equation

domain

collapse

№35 Cosine Knows No Sign

● ● ● ● ● L3

SIMPLIFY

Simplify $g(x) = \cos^{-1}\left(\frac{1-x^2}{1+x^2}\right)$ for all real x , giving the piecewise form.

cosine-double-angle

arctangent

absolute-value

collapse

№36 Two Faces of 2arctan

●●●●● L4

PROVE THAT

Prove that for $x > 1$, $2 \tan^{-1} x = \pi - \sin^{-1} \left(\frac{2x}{1+x^2} \right)$.

arctangent

sine-form

branch

collapse

№37 The Slope That Flips

●●●●● L4

DIFFERENTIATE

For $f(x) = \sin^{-1} \left(2x \sqrt{1-x^2} \right)$, find $f'(x)$ at $x = \frac{\sqrt{3}}{2}$.

sine-double-angle

derivative

branch

collapse

№38 Counting the Honest x

●●●●● L3

FIND THE NUMBER OF SOLUTIONS

Find the number of integers $x \in [-5, 5]$ satisfying $\cos^{-1} \left(\frac{1-x^2}{1+x^2} \right) = 2 \tan^{-1} x$.

cosine-double-angle

arctangent

integer-count

collapse

№39 A Telescope in Disguise

●●●●● L4

EVALUATE

Evaluate $\sum_{k=1}^4 \tan^{-1} \left(\frac{1}{1+k+k^2} \right)$.

arctangent

difference-formula

telescoping

collapse

№40 The Tent's Full Reach

●●●●● L3

FIND THE RANGE

Find the range of $f(x) = \sin^{-1} \left(2x \sqrt{1-x^2} \right)$ over its natural domain $[-1, 1]$.

sine-double-angle

range

piecewise

collapse

CHAPTER V

Equations & Root-Counting

Every inverse-trig equation is a domain problem in disguise.

№41 A Tilted Pair of Sines

● ● ● ● ● L4

SOLVE FOR X

$$\sin^{-1} x + \sin^{-1}(2x) = \frac{\pi}{3}$$

arcsin

squaring

extraneous-root

principal-range

№42 Two Arctangents, One Target

● ● ● ● ● L4

SOLVE FOR X

$$\tan^{-1}(x+1) + \tan^{-1}(x-1) = \tan^{-1}\left(\frac{8}{31}\right)$$

arctan

addition-formula

ab-less-than-one

extraneous-root

№43 Where Arccos Meets Arctan

● ● ● ● ● L4

FIND THE NUMBER OF SOLUTIONS

Find the number of real solutions of $\cos^{-1} x = \tan^{-1} x$.

arccos

arctan

range-intersection

monotonicity

№44 The Half-Angle Disguise

● ● ● ● ● L4

SOLVE FOR X

$$\tan^{-1}\left(\frac{1-x}{1+x}\right) = \frac{1}{2} \tan^{-1} x, \quad x > 0$$

arctan

subtraction-formula

half-angle

domain-validity

№45 A Sine Splits in Two

● ● ● ● ● L4

SOLVE FOR X

$$\sin^{-1} x + \sin^{-1}(1-x) = \cos^{-1} x$$

arcsin

arccos

double-angle

complementary-identity

№46 Cotangent Minus Tangent

●●●●● L4

SOLVE FOR X

$$\cot^{-1} x - \tan^{-1} x = \frac{\pi}{6}$$

arccot

arctan

complementary-identity

principal-branch

№47 The Vanishing Radical

●●●●● L4

SOLVE FOR X

$$\tan^{-1} \sqrt{x^2 + x} + \sin^{-1} \sqrt{x^2 + x + 1} = \frac{\pi}{2}$$

arctan

arcsin

domain-pinching

radical

№48 Reciprocal Twins

●●●●● L4

FIND THE NUMBER OF SOLUTIONS

Find the number of real solutions of $\tan^{-1}\left(\frac{1}{2x-1}\right) + \tan^{-1}\left(\frac{1}{2x+1}\right) = \tan^{-1}\left(\frac{4}{7}\right).$

arctan

addition-formula

ab-less-than-one

root-counting

№49 The Triple-Angle Plateau

●●●●● L5

FIND THE NUMBER OF SOLUTIONS

Find the number of real solutions of $3 \cos^{-1} x - \cos^{-1}(4x^3 - 3x) = \pi.$

arccos

triple-angle

branch-piecewise

root-counting

№50 The Sum of Squared Angles

●●●●● L5

SOLVE FOR X

$$(\sin^{-1} x)^2 + (\cos^{-1} x)^2 = \frac{5\pi^2}{8}$$

arcsin

arccos

sum-pi-over-2

range-rejection

CHAPTER VI

Trig of Inverse Trig

The trig of a sum of inverse functions — drawn on a triangle.



№51 The Three-Step Collapse

● ● ● ● ● L3

EVALUATE

$$\tan\left(2 \tan^{-1} \frac{1}{3} + \tan^{-1} \frac{1}{7}\right)$$

tangent

double-angle

addition-formula

right-triangle

№52 When the Obtuse Angle Cancels

● ● ● ● ● L3

EVALUATE

$$\sin\left(\cos^{-1}\left(-\frac{4}{5}\right) + \tan^{-1} \frac{3}{4}\right)$$

sine

addition-formula

obtuse-quadrant

sign-of-root

№53 Half of an Obtuse Angle

● ● ● ● ● L4

EVALUATE

$$\tan\left(\frac{1}{2} \cos^{-1}\left(-\frac{7}{25}\right)\right)$$

half-angle

tangent

obtuse-quadrant

sign-of-root

№54 A Difference of Two Signs

● ● ● ● ● L4

EVALUATE

$$\sin\left(\cos^{-1}\left(-\frac{5}{13}\right) - \sin^{-1}\left(-\frac{3}{5}\right)\right)$$

sine

difference-formula

two-quadrants

sign-of-root

№55 Doubling a Sharp Angle

● ● ● ● ● L3

EVALUATE

$$\cos\left(2 \cos^{-1} \frac{1}{4}\right)$$

cosine

double-angle

right-triangle

rational-value

№56 The Surd in the Denominator

●●●●● L4

EVALUATE

$$\tan\left(2\cos^{-1}\left(-\frac{3}{\sqrt{13}}\right)\right)$$

tangent double-angle obtuse-quadrant surd-input sign-of-root

№57 Secant Meets Cosecant

●●●●● L4

EVALUATE

$$\sin\left(\sec^{-1}\left(-\frac{13}{5}\right) + \csc^{-1}\frac{5}{3}\right)$$

sine addition-formula secant-inverse cosecant-inverse sign-of-root

№58 The Symbolic Mirror

●●●●● L3

SIMPLIFY

$$\cos(\sin^{-1}x - \cos^{-1}x) \quad \text{for } x \in [-1, 1]$$

cosine symbolic complementary-identity double-angle

№59 Halving a Near-Straight Angle

●●●●● L4

EVALUATE

$$\sin\left(\frac{1}{2}\cos^{-1}\left(-\frac{119}{169}\right)\right)$$

half-angle sine obtuse-quadrant sign-of-root pythagorean-triple

№60 The Sum That Overshoots

●●●●● L5

EVALUATE

$$\sin(\tan^{-1}2 + \tan^{-1}3)$$

sine tangent-addition branch-overflow principal-value

CHAPTER VII

Domain, Range & Graphs



Where is the composite defined, and how often does its graph cross?

№61 A Logarithm Inside the Cosine

● ● ● ● ● L3

FIND THE DOMAIN

$$f(x) = \cos^{-1}(\log_2 x)$$

domain

composite

logarithm

cosine-inverse

№62 Sine-Inverse Meets a Logarithmic Tail

● ● ● ● ● L3

FIND THE DOMAIN

$$f(x) = \sin^{-1}\left(\frac{x-3}{2}\right) + \ln(4-x)$$

domain

sine-inverse

logarithm

intersection

№63 Two Increasing Arcs Stacked

● ● ● ● ● L4

FIND THE RANGE

$$f(x) = \sin^{-1} x + \tan^{-1} x$$

range

monotonic

sine-inverse

tangent-inverse

№64 A Weighted Difference of Arcs

● ● ● ● ● L4

FIND THE RANGE

$$f(x) = 2 \cos^{-1} x - \sin^{-1} x$$

range

monotonic

cosine-inverse

sine-inverse

weighted

№65 Counting Lattice Points in a Split Domain

● ● ● ● ● L4

FIND THE NUMBER OF INTEGERS IN THE DOMAIN

$$f(x) = \sin^{-1}\left(\log_3 \frac{x^2}{3}\right)$$

domain

integer-count

logarithm

disconnected

sine-inverse

№66 A Circle Caps the Arcsine

●●●●● L4

FIND THE NUMBER OF INTEGERS IN THE DOMAIN

$$f(x) = \sin^{-1}\left(\frac{|x| - 2}{3}\right) + \sqrt{16 - x^2}$$

domain integer-count square-root absolute-value intersection

№67 Sum of Squared Arcs

●●●●● L4

FIND THE RANGE

$$f(x) = (\sin^{-1} x)^2 + (\cos^{-1} x)^2$$

range quadratic sine-inverse cosine-inverse optimization

№68 The Branch-Folded Constant

●●●●● L4

FIND THE RANGE

$$f(x) = 2 \tan^{-1} x + \sin^{-1}\left(\frac{2x}{1+x^2}\right)$$

range branch double-angle tangent-inverse piecewise

№69 When the Secant Argument Never Fails

●●●●● L5

FIND THE DOMAIN

$$f(x) = \sec^{-1}\left(\frac{x^2 + 3}{2x}\right)$$

domain secant-inverse rational am-gm absolute-value

№70 A Gentle Line Through the Sawtooth

●●●●● L3

FIND THE NUMBER OF SOLUTIONS

$$\sin^{-1}(\sin x) = \frac{x}{10}$$

graph sawtooth intersection sine-inverse periodic

CHAPTER VIII

The Calculus of Inverse Trig

d/dx

Differentiate across a branch point and the chain rule lies by a sign.

№71 The Phantom Jump

●●●●● L4

DIFFERENTIATE

Let $f(x) = \tan^{-1}\left(\frac{2x}{1-x^2}\right)$. Find $f'(x)$ for all $x \neq \pm 1$, and decide whether f is differentiable at $x = 2$.

derivative

branch

tangent-double-angle

discontinuity

№72 Sawtooth Slopes

●●●●● L4

EVALUATE

Let $g(x) = \sin^{-1}(\sin x)$. Evaluate $g'(2) + g'(5) + g'(8)$.

sawtooth

sin-inverse

piecewise

derivative-at-point

№73 Two Faces of a Double Angle

●●●●● L4

DIFFERENTIATE

Let $h(x) = \sin^{-1}\left(\frac{2x}{1+x^2}\right)$. Find $h'(x)$ for $|x| < 1$ and for $|x| > 1$, and state $h'\left(\frac{1}{\sqrt{3}}\right)$ and $h'(\sqrt{3})$.

sin-inverse

double-angle

branch-sign

piecewise

№74 A Cubic Whisper

●●●●● L4

EVALUATE

$$\lim_{x \rightarrow 0} \frac{\tan^{-1} x - \sin^{-1} x}{x^3}.$$

limit

taylor-series

tan-inverse

sin-inverse

№75 The Tilted Tangent

●●●●● L4

DIFFERENTIATE

For $x \in (0, \frac{3\pi}{4})$ let $F(x) = \tan^{-1}\left(\frac{\cos x - \sin x}{\cos x + \sin x}\right)$. Find $F'(x)$, and find $F'(x)$ for $x \in (\frac{3\pi}{4}, \frac{7\pi}{4})$.

tan-inverse

reduction

derivative-at-point

branch

№76 Half the Angle, Half the Slope

●●●●● L4

DIFFERENTIATE

For $x \in (0, \pi)$ let $P(x) = \tan^{-1} \sqrt{\frac{1 - \cos x}{1 + \cos x}}$. Find $P'(x)$, and find $Q'(x)$ where $Q(x) = \tan^{-1} \sqrt{\frac{1 - \cos x}{1 + \cos x}}$ on $(\pi, 2\pi)$.

half-angle

tan-inverse

domain

derivative-at-point

№77 The Triple-Angle Mirage

●●●●● L5

DIFFERENTIATE

Let $S(x) = \sin^{-1}(3x - 4x^3)$, $x \in [-1, 1]$. Find $S'(x)$ on each subinterval determined by $x = \pm \frac{1}{2}$, and give $S'(\frac{1}{4})$ and $S'(\frac{3}{4})$.

triple-angle

sin-inverse

domain

branch-sign

piecewise

№78 Integrating the Sawtooth

●●●●● L5

EVALUATE

$$\int_0^{2\pi} \sin^{-1}(\sin x) dx \quad \text{and} \quad \int_0^{\pi} \sin^{-1}(\sin x) dx.$$

definite-integral

sawtooth

sin-inverse

piecewise

symmetry

№79 Two Sawtooths Collide

●●●●● L4

EVALUATE

Let $W(x) = \sin^{-1}(\sin x) + \cos^{-1}(\cos x)$. Evaluate $W'(\frac{\pi}{3})$ and $W'(2)$.

sawtooth

sin-inverse

cos-inverse

derivative-at-point

piecewise

№80 The Even Derivative

●●●●● L4

DIFFERENTIATE

Let $R(x) = \sec^{-1} x$, $|x| \geq 1$. Find $R'(2)$ and $R'(-2)$, and state the sign relationship between them.

sec-inverse

absolute-value

derivative-at-point

branch

CHAPTER IX

Machin & Olympiad Identities



Machin's π , Gaussian-integer arctangents, the competition canon.

№81 Hutton's Hidden Twin

● ● ● ● ● L4

PROVE THAT

$$2 \tan^{-1} \frac{1}{3} + \tan^{-1} \frac{1}{7} = \frac{\pi}{4}$$

machin gaussian-integer arctan-sum principal-value

№82 The Sum That Refuses to Vanish

● ● ● ● ● L4

PROVE THAT

$$\tan^{-1} 1 + \tan^{-1} 2 + \tan^{-1} 3 = \pi$$

arctan-sum branch gaussian-integer quadrant

№83 Telescope to a Quarter Turn

● ● ● ● ● L4

EVALUATE

$$\sum_{n=1}^{\infty} \tan^{-1} \left(\frac{1}{n^2 + n + 1} \right)$$

telescoping series arctan-difference limit

№84 Hutton's Two-Term Machine

● ● ● ● ● L4

PROVE THAT

$$2 \tan^{-1} \frac{1}{2} - \tan^{-1} \frac{1}{7} = \frac{\pi}{4}$$

machin gaussian-integer conjugate arctan-difference

№85 Where Cosine Meets Cosecant

● ● ● ● ● L5

SOLVE FOR X

$$2 \tan^{-1}(\cos x) = \tan^{-1}(2 \csc x), \quad x \in (0, \pi)$$

equation domain principal-value trig-substitution

№86 Euler's Three Unit Fractions

●●●●● L4

PROVE THAT

$$\tan^{-1} \frac{1}{2} + \tan^{-1} \frac{1}{5} + \tan^{-1} \frac{1}{8} = \frac{\pi}{4}$$

arctan-sum

gaussian-integer

euler

principal-value

№87 The Obtuse Sum

●●●●● L3

EVALUATE

$$\tan^{-1} 2 + \tan^{-1} 3$$

arctan-sum

branch

quadrant

tangent-addition

№88 A Right Angle in Three Pieces

●●●●● L4

PROVE THAT

$$\tan^{-1} 1 + \tan^{-1} \frac{1}{2} + \tan^{-1} \frac{1}{3} = \frac{\pi}{2}$$

arctan-sum

gaussian-integer

pure-imaginary

euler

№89 Stormer's Hat-Trick

●●●●● L5

PROVE THAT

$$6 \tan^{-1} \frac{1}{8} + 2 \tan^{-1} \frac{1}{57} + \tan^{-1} \frac{1}{239} = \frac{\pi}{4}$$

machin

gaussian-integer

high-multiplicity

239

№90 The Skipping Telescope

●●●●● L5

EVALUATE

$$\sum_{n=1}^{\infty} \tan^{-1} \left(\frac{2}{n^2} \right)$$

telescoping

series

arctan-difference

two-step

CHAPTER X

The Grand Hybrids

Branch correction, telescoping, equations and calculus, fused.



№91 The Gap-Two Telescope That Overshoots

● ● ● ● ● L5

EVALUATE

$$\cos^{-1}\left(\cos\left(2\sum_{k=1}^{\infty}\tan^{-1}\frac{2}{k^2}\right)\right)$$

telescoping

arctan

cos-inverse-cos

branch

series

№92 Half-Angle Trap on the Möbius Argument

● ● ● ● ● L4

SOLVE FOR X

$$\tan^{-1}\left(\frac{1-x}{1+x}\right) = \frac{1}{2}\tan^{-1}x$$

arctan

möbius-argument

branch

root-count

identity-collapse

№93 Differentiating Across the Fold

● ● ● ● ● L5

DIFFERENTIATE

$$\text{Let } f(x) = \sin^{-1}(2x\sqrt{1-x^2}). \text{ Find } f'\left(\frac{\sqrt{3}}{2}\right).$$

derivative

double-angle

branch

sin-inverse

point-evaluation

№94 The Pi-Chain Plus a Sixth

● ● ● ● ● L5

EVALUATE

$$\cos^{-1}\left(\cos\left(\tan^{-1}1 + \tan^{-1}2 + \tan^{-1}3 + \cos^{-1}\left(-\frac{1}{2}\right)\right)\right)$$

arctan-chain

gaussian-integers

cos-inverse-cos

branch

reduction

№95 Two Sines Equal a Cosine

● ● ● ● ● L4

SOLVE FOR X

$$\sin^{-1}x + \sin^{-1}(1-x) = \cos^{-1}x$$

sin-inverse

cos-inverse

identity-collapse

root-count

domain

№96 A One-Sided Slope at the Origin

●●●●● L5

EVALUATE

$$\lim_{x \rightarrow 0^-} \frac{\cos^{-1}\left(\frac{1-x^2}{1+x^2}\right)}{x}$$

limit cos-inverse double-angle branch one-sided

№97 Tripling the Three-Term Chain

●●●●● L5

EVALUATE

$$\cos^{-1}\left(\cos\left(3\tan^{-1}1 + 3\tan^{-1}2 + 3\tan^{-1}3\right)\right)$$

arctan-chain cos-inverse-cos branch reduction gaussian-integers

№98 Sum of Squares Meets the Range Filter

●●●●● L5

SOLVE FOR X

$$(\sin^{-1}x)^2 + (\cos^{-1}x)^2 = \frac{5\pi^2}{8}$$

sin-inverse cos-inverse quadratic range-filter root-count

№99 The Rotated Tangent, Read Far Right

●●●●● L4

EVALUATE

$$\text{Let } h(x) = \tan^{-1}\left(\frac{\cos x - \sin x}{\cos x + \sin x}\right). \text{ Find } h(3).$$

tan-inverse-tan phase-shift branch point-evaluation piecewise

№100 Three Undoings at Five Radians

●●●●● L3

EVALUATE

$$\sin^{-1}(\sin 5) + \cos^{-1}(\cos 5) + \tan^{-1}(\tan 5)$$

sin-inverse-sin cos-inverse-cos tan-inverse-tan quadrant reduction